Superpixel Contracted Neighborhood Contrastive Subspace Clustering Network for Hyperspectral Images

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***Abstract*— Deep subspace clustering (DSC) has achieved remarkable performances in the unsupervised classification of hyperspectral images. However, previous models based on pixel-level self-expressiveness of data suffer from the exponential growth of computational complexity and access memory requirements with an increasing number of samples, thus leading to poor applicability to large hyperspectral images. This article presents a neighborhood contrastive subspace clustering (NCSC) network, a scalable and robust DSC approach, for unsupervised classification of large hyperspectral images. Instead of using a conventional autoencoder, we devise a novel superpixel pooling autoencoder to learn the superpixel-level latent representation and subspace, allowing a contracted self-expressive layer. To encourage a robust subspace representation, we propose a novel neighborhood contrastive regularization to maximize the agreement between positive samples in subspace. We jointly train the resulting model in an end-to-end fashion by optimizing an adaptively weighted multitask loss. Extensive experiments on three hyperspectral benchmarks demonstrate the effectiveness of the proposed approach and its substantial advancement of state-of-the-art approaches.**

***Index Terms*— Contrastive learning, hyperspectral image processing, subspace clustering, superpixel.**

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I. Introduction

HYPERSPECTRAL imaging techniques have facilitated numerous promising application scenarios, ranging from ocean, urban, agriculture, and geology to biomedicine [1], [2], due to its unique advantages. Hyperspectral imagery (HSI) acquired by imaging spectrometer often consists of dozens of spectral bands with a few nanometers spectral resolution. Such rich spectral information enables objects of interest to be classified at fine-grained level [3], [4].

Despite this tremendous success, supervised HSI classification techniques [5], [6] largely rely on the availability of large amounts of annotated data, which require intensive manual labor. To avoid the burden of human annotation, the remote sensing community has shifted its attention to techniques that can learn with nonannotated data [7]. To this end, increasing numbers of clustering methods have been proposed, narrowing the gap in accuracy between unsupervised and supervised methods. Nonetheless, HSI clustering is still a fundamental but challenging task because of the complexity of HSI data, especially the large spectral variability of the targets [8].

Over the past few decades, subspace-based methods [9], [10] have enabled significant progress in clustering high-dimensional HSI by assuming that datasets are distributed around a collection of linear and affine subspaces. Despite promising performances, traditional subspace clustering implies self-expression that requires high computation and memory facilities and limits its applicability to small-scale problems. As a temporary compromise, most previous works chose to evaluate their models on small HSI datasets [11]–[15]. Recent efforts in the machine learning community have been devoted to scaling-up subspace clustering to large-scale datasets. Although most of them were not initially proposed for HSI, they are adapted to overcome this limitation. The most popular technique is representative points sampling [16]–[18]. The key idea is to select a representative subset of samples, which is often termed landmarks [18], [19], exemplars [20], or dictionary [21], according to a given rule. For example, the representative subset is desired to cover all potential subspaces, on which the subspace learning can be easily conducted. Another candidate to handle large HSI is to use superpixel segmentation [22], [23] by considering spatial homogeneity.

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However, these methods follow a linear and shallow subspace assumption, which does not hold for real-world HSI data.

In contrast, deep subspace clustering (DSC) [24]–[26] incorporates the self-expressiveness property of data with a deep autoencoder, making it possible to exploit high-level features and nonlinear subspace. Due to the powerful ability of representation learning, DSC can fully leverage the rich spectral and spatial information found in HSI [15], [27], thus resulting in more accurate subspace segmentation. However, DSC suffers from several drawbacks when dealing with real HSI. First, the scalable techniques used in classic subspace clustering models fail to work in the self-expressive (SE) layer of the DSC model, leading to a critical bottleneck that limits its handling of large HSI. Second, DSC attempts to generate feature representations and cluster assignments at the pixel level, which not only wastes substantial computing resources but also ignores the essential homogeneousness of HSI. Third, the absence of robust regularization on self-expression coefficients gives DSC poor robustness to the noise that commonly affects HSI, as well as slow convergence speed. Fourth, the training of DSC derives a typical multitask learning problem that is strongly dependent on the relative weighting between each task's loss; however, tuning these weights by hand is a difficult and time arduous process, making it prohibitive in practice.

In this article, we address these drawbacks by introducing a superpixel contracted neighborhood contrastive subspace clustering (NCSC) network. Specifically, we improve the previous DSC in three main respects. First, instead of adopting a pixel-level autoencoder, we design a novel superpixel pooling autoencoder (SPAE) to learn superpixel-level deep self-expression in an end-to-end manner. Since there are far fewer superpixels than pixels, SPAE significantly reduces the complexity of the SE layer, thereby enabling clustering on large HSI. Second, we introduce neighborhood contrastive regularization (NCR) to enhance the robustness of subspace learning. The idea is to pull together positive samples indicated by neighborhood while pushing away negative samples in underlying subspaces. Third, we adaptively weight multiple losses by predicting the variance of each loss, which makes model training moderate in practice.

Our contributions are summarized as follows.

1) By leveraging the homogeneousness of HSI, we scale up DSC to large HSI datasets through the use of superpixel pooling (SP). The proposed NCSC is a holistic approach, designed to meet triple objectives: global pixel reconstruction, local superpixel reconstruction, and nonlocal neighborhood contrastive learning.
2) We ease the training of DSC by introducing a robust regularization, i.e., NCR, and adopting an adaptive multitask loss weighting, both of which significantly facilitate subspace learning and model convergence.
3) We systematically evaluate our approach on highly competitive HSI benchmarks, demonstrating that the proposed approach is an improvement over state-of-the-art approaches by a nontrivial margin.

The rest of this article is structured as follows. Section II gives a brief review of subspace clustering algorithms and HSI clustering approaches. In Section III, we describe the details of the proposed NCSC approach, including SPAE, NCR, and training details. In Section IV, we provide systematic experimental results for our approach on several real HSI datasets. Finally, we conclude with a summary and future works in Section V.

II. Related Work

A. Subspace Clustering

Let $\mathcal{X} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N\} \in \mathbb{R}^D$ be a collection of N data points drawn from the D-dimensional space and belonging to C distinct clusters. Subspace clustering [9] assumes that $\mathcal{X}$ lies in a union of the low-dimensional linear subspace $\mathcal{R}_1 \cup \mathcal{R}_2 \cup \cdots \cup \mathcal{R}_C$, where each $\boldsymbol{x}_i$ can be expressed by a linear combination of the data points belonging to the same subspace. This assumption is referred to as the self-expressiveness property of data [10]. Formally, subspace clustering solves the following optimization problem:

$$\arg\min_{\mathbf{Z}} \frac{1}{2}\|\mathbf{X} - \mathbf{Z}\mathbf{X}\|_F^2 + \lambda\|\mathbf{Z}\|_p, \quad \text{s.t. } \operatorname{diag}(\mathbf{Z}) = 0. \tag{1}$$

Here, $\mathbf{X} \in \mathbb{R}^{N\times D}$ is the matrix form of the data points, $\mathbf{Z} \in \mathbb{R}^{N\times N}$ denotes the self-expression coefficient matrix, and λ indicates a regularization coefficient. In particular, different forms of matrix norm $\|\cdot\|_p$ will derive variants of subspace clustering. For example, the classic sparse subspace clustering (SSC) [10], low-rank representation (LRR) [28], and least-squares regression (LSR) [29] adopt the ℓ_1 norm, nuclear norm, and Frobenius norm, respectively. The self-expression coefficient matrix is usually subspace-preserving in contrast to the pairwise similarity matrix. Such property reveals the intrinsic structure of the data, thus being more robust and interpretable.

In recent years, many advanced subspace clustering models have been proposed to address specific problems. On the one hand, a variety of priors can be considered during the subspace clustering. For example, the structural information of the data is a type of high-order prior that has been widely used for manifold regularization or Laplacian regularization [27], [30]. On the other hand, DSC [24], [25] not only has significantly improved classic subspace clustering but also has facilitated the design and application of learning models. Instead of implicitly exploiting nonlinear relationships such as kernel subspace clustering [31], DSC explicitly parameterizes the nonlinear latent space using a neural network, resulting in a more flexible learning framework.

B. HSI Clustering Methods

Given an HSI cube $\mathbf{X} \in \mathbb{R}^{W\times H\times D}$, which consists of $N = W \times H$ pixels and D spectral bands, HSI clustering aims to group every pixel $\mathcal{X} = \{\boldsymbol{x}_i \in \mathbb{R}^D\}_{i=1}^{W\times H}$ into C distinct clusters such that spectral or spatial patterns within the same clusters are similar to each other, while those in different clusters are dissimilar. Therefore, HSI clustering is a typical combinatorial optimization problem. We broadly divide existing HSI clustering approaches into two categories: classic- and representation-based approaches. The classic approach

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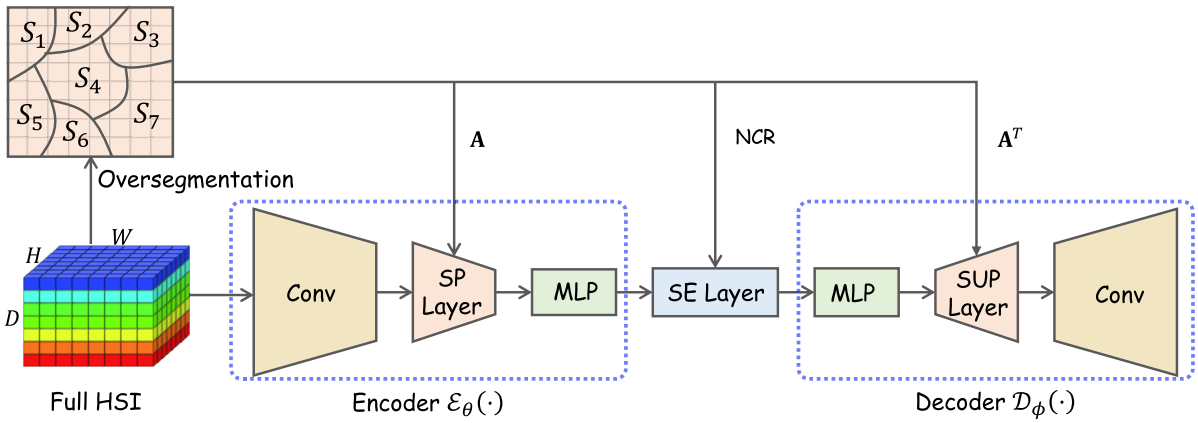


Fig. 1. Overview of the proposed NCSC approach. Here, we use an SPAE to jointly learn a robust superpixel-level self-expression coefficient matrix with regularization of NCR. The forward phase of the model takes the full HSI $\mathbf{X} \in \mathbb{R}^{W\times H\times B}$ and an association matrix $\mathbf{A}$ as inputs and calculates the superpixel-level representation by encoder $\mathcal{E}_\theta(\cdot)$ and pixel-level reconstruction by decoder $\mathcal{D}_\phi(\cdot)$. The intermediate SE layer conducts a superpixel-level reconstruction, which aims to learn a contracted subspace representation. NCR: neighborhood contrastive regularization. SP: superpixel pooling. SUP: superpixel unpooling. MLP: multilayer perceptron. SE: self-expressive layer.

can be further subdivided into centroid-based methods, such as k-Means [32] and fuzzy c-Means (FCM) [33], graph-based methods, e.g., spectral clustering [19], and so on. The representation-based methods mainly include subspace clustering and deep clustering. Due to its excellent performance, the representation-based method has become the most popular choice in HSI clustering.

Early HSI clustering methods focus mainly on the feasibility of applying subspace clustering to HSI [12]–[15], [34]. Despite the promising performance, these works are highly limited in their applicability to large HSI scenes due to pixel-level self-expression. More recently, an increasing number of works have shifted their attention to handling large HSI datasets. Successful attempts can be summarized into three types: representative points sampling [17], [35], mini-batch training [36], [37], and superpixel segmentation [22]. As introduced in Section I, the representative points sampling-based methods warrant the obtention of a representative subset from the whole dataset. However, the subset relies heavily on heuristic searching. In particular, it is difficult to use the representative points sampling in a DSC network. Mini-batch training is widely used in some deep clustering models, such as deep embedding clustering (DEC) [37]. Nonetheless, such a training manner does not apply to DSC networks because the SE layer requires a full batch input. Superpixel segmentation [6], [38], [39] has some advantages, including effectively reducing computational complexity and preserving locality and homogeneity. Surprisingly, there are only few works describing the upscaling of DSC using superpixels. Based on this insight, we propose NCSC, which can be seen as a superpixel-based scalable approach.

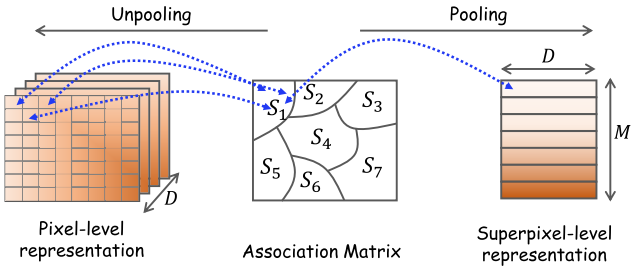


Fig. 2. Illustration of SP and unpooling operations. We implement these operations by using an association matrix that indexes pixel-to-superpixel membership. In the text, superpixel pooling and superpixel unpooling are abbreviated as SP and SUP, respectively.

III. Proposed Method

A. Overall Architecture

The overall architecture of the proposed NCSC approach is shown in Fig. 1. Similar to existing DSC models [15], [24], [27], we recast the classic subspace clustering into an SE layer of an autoencoder to learn the nonlinear subspace representation in a latent space. However, instead of using a standard pixel-level autoencoder, we propose a novel SPAE to efficiently transform between pixel-level representations and superpixel-level representations. To this end, we introduce an SP layer and superpixel unpooling (SUP) layer into the encoder and decoder of SPAE, respectively. This allows us to compute a contracted self-expression in a superpixel-level latent space, instead of using the costly pixel-level representation. Formally, we denote the encoder as $\mathbf{H} = \mathcal{E}_\theta(\cdot)$, which produces the superpixel-level representation $\mathbf{H}$ with trainable parameters θ, and the decoder as $\widehat{\mathbf{X}} = \mathcal{D}_\phi(\cdot)$, which seeks to reconstruct inputs with trainable parameters ϕ. We introduce the SE layer between the encoder and the decoder. As a result, the overall forward phase of the resulting network can be expressed as $\widehat{\mathbf{X}} = \mathcal{D}_\phi(\mathbf{Z}\mathcal{E}_\theta(\mathbf{X}))$, where $\mathbf{Z}$ is the superpixel-level subspace representation coefficient matrix. Our goal is to learn a robust coefficient matrix by jointly training the network with suitable regularizations. When the model training is complete, we apply spectral clustering on the optimal affinity matrix $\mathbf{W} = (1/2)(|\mathbf{Z}^{\mathrm{T}}| + |\mathbf{Z}|)$ to perform the subspace segmentation. More details of our approach are provided in the following.

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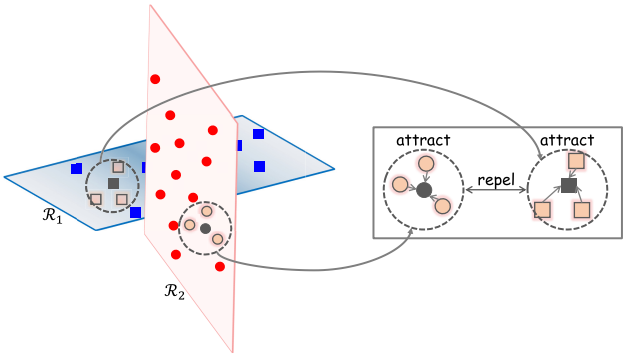


Fig. 3. Graphical representation of NCR on subspaces $\mathcal{R}_1$ and $\mathcal{R}_2$. The gray points denote anchors and their positive samples are marked in orange. NCR will attract positive samples and repel negative samples.

B. Generation of Homogeneous Regions

Based on the property of natural images, the pixels belonging to the same object often are represented as homogeneous regions at the semantic level. This enables us to convert the original pixel-level HSI clustering problem into a superpixel-level problem. Specifically, we denote the M nonoverlapping homogeneous regions of an HSI as $\mathcal{S} = \{\mathcal{S}_i; |\mathcal{S}_i| = n_i\}_{i=1}^{M}$, where $M \ll N$ and each $\mathcal{S}_i$ contains n_i pixels. The original SE layer with the complexity of $\mathcal{O}(N^2)$ will be reduced to $\mathcal{O}(M^2)$, thus making it scalable to large HSI. Furthermore, the use of homogeneous regions may guarantee the semantic consistency of clustering results.

In this article, we use entropy rate superpixel segmentation (ERS) [40] to generate $\mathcal{S}$ due to its superiority in efficiency and efficacy [39]. It should be noted that the choice of M is an open problem. Here, we follow [39] and adaptively set $M = \alpha(N_{\text{edge}}/N)$, where N_{edge} denotes the number of nonzero elements in the detected edge image[1] of an HSI and α is a fixed base number for all HSIs. In our experiments, we empirically set $\alpha = 3000$.

C. Superpixel Pooling Autoencoder

Inspired by the SP network adopted in semantic segmentation [38], [41], we propose an SPAE to learn superpixel-level feature representations. Compared to the previous SP network [38], our SPAE is unsupervised and explicitly adopts an inverses operation called SUP. Specifically, our SPAE consists of an encoder–decoder architecture comprising multiple optional convolutional and linear layers. The major difference between SPAE and traditional AE is the SP/SUP layer. In Fig. 2, we show how the SP and SUP operations work. Similar to conventional average pooling operations, SP downsamples pixel-level feature representation with the size of N into M single vectors by averaging within each homogeneous region. Let $\mathbf{H}_P \in \mathbb{R}^{N\times D}$ and $\mathbf{H}_S \in \mathbb{R}^{M\times D}$ be the pixel-level and superpixel-level representations, respectively. The SP operation can be mathematically denoted as the following linear transformation:

$$\mathbf{H}_S = \mathbf{A}\mathbf{H}_P. \tag{2}$$

Here, $\mathbf{A} \in \mathbb{R}^{M\times N}$ indicates an association matrix, where $A_{ij} = \mathbb{1}(\boldsymbol{x}_j \in \mathcal{S}_i)$, signifying whether pixel $\boldsymbol{x}_j$ belongs to the superpixel $\mathcal{S}_i$. In contrast, SUP reverses the SP operation by upsampling the superpixel-level representations to the initial size. The process can be written as

$$\widehat{\mathbf{H}}_P = \mathbf{A}^{\mathrm{T}}\mathbf{H}_S. \tag{3}$$

The SUP operation is equivalent to repeating each superpixel-level representation n_i times to fill $\mathcal{S}_i$. We use the SP layer and SUP layer in the encoder and decoder, respectively. Since both SP and SUP may generate coarse feature representations, we adopt additional MLPs to refine these features.

To exploit the spatial information, we adopt patch-like inputs for SPAE and extract spectral–spatial information using convolution layers. In the decoder, due to the good balance between computational complexity and upsampling quality, we use the bilinear interpolation instead of deconvolution to rescale the downsampled feature maps to their initial size. The objective function of SPAE is defined as the square error between the initial inputs $\mathbf{X}$ and the final reconstructions $\widehat{\mathbf{X}}$ denoted as

$$\mathcal{L}_{\text{AE}} = \|\mathbf{X} - \mathcal{D}_\phi(\mathcal{E}_\theta(\mathbf{X}))\|_F^2 = \|\mathbf{X} - \widehat{\mathbf{X}}\|_F^2. \tag{4}$$

The reconstruction of inputs is approximated by minimizing the objective function.

D. Superpixel Contracted Self-Expression

We use the SE layer to explore the self-expressiveness of an HSI in the nonlinear latent space parameterized by SPAE. Since there are far fewer superpixels than initial pixels, the complexity of the superpixel-level SE layer is always limited to data amount suitable to its computation using standard equipment and not an high-performance computer (HPC). Similar to (1), we denote the SE layer as the following linear transformation with an identity constraint:

$$\widehat{\mathbf{H}} = \mathbf{Z}\mathbf{H}, \quad \text{s.t. } \widehat{\mathbf{H}} = \mathbf{H}. \tag{5}$$

This transformation is achieved by minimizing the following objective function:

$$\mathcal{L}_{\text{SE}} = \|\mathbf{Z}\mathbf{H} - \mathbf{H}\|_F^2. \tag{6}$$

This objective function allows the SE layer to be jointly trained with the whole SPAE network in an end-to-end manner. However, because of the lack of a strict constraint, the objective function may lead to a trivial solution, i.e., $\mathbf{Z} = \mathbf{I}$. ℓ_1 norm, ℓ_2 norm, and graph regularization (GR) have been considered in previous works [24], [27] to overcome similar problems. As an alternative to these existing regularization approaches, in Section III-E, we show that they are not adapted to the proposed approach and introduce a novel regularization method for the SE layer.

[1]The edge image can be obtained by any edge detection algorithm. Here, we use the build-in edge function of MATLAB.

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TABLE I

GROUND TRUTH CLASSES FOR THE TRENTO, SALINAS, AND INDIAN PINES 2010 DATASETS AND THEIR RESPECTIVE SAMPLES NUMBER

Class No.	Trento		Salinas		Indian Pines 2010	
	Class Name	# Samples	Class Name	# Samples	Class Name	#Samples
1	Apple trees	4034	Brocoli_green_weeds_1	2009	Corn-high	3387
2	Buildings	2903	Brocoli_green_weeds_2	3726	Corn-mid	1740
3	Ground	479	Fallow	1976	Corn-low	356
4	Wood	9123	Fallow_rough_plow	1394	Soy-bean-high	1365
5	Vineyard	10501	Fallow_smooth	2678	Soy-bean-mid	37865
6	Roads	3174	Stubble	3959	Soy-bean-low	29210
7			Celery	3579	Residues	5795
8			Grapes_untrained	11271	Wheat	3387
9			Soil_vinyard_develop	6203	Hay	50045
10			Corn_senesced_green_weeds	3278	Grass/Pasture	5544
11			Lettuce_romaine_4wk	1068	Cover crop 1	2746
12			Lettuce_romaine_5wk	1927	Cover crop 2	2164
13			Lettuce_romaine_6wk	916	Woodlands	48559
14			Lettuce_romaine_7wk	1070	Highway	4863
15			Vinyard_untrained	7268	Local road	502
16			Vinyard_vertical_trellis	1807	Buildings	546
Total	N/A	30214	N/A	54129	N/A	198074

TABLE II

NETWORK DETAILS OF THE PROPOSED NCSC METHOD

		Layer	Output Size	Kernel
Encoder	Conv	Conv1	$N \times 64 \times 7 \times 7$	$D \times 64 \times 1 \times 1$
		Conv2	$N \times 64 \times 5 \times 5$	$64 \times 64 \times 3 \times 3$
		Avg Pool	$N \times 64$	–
		Linear1	$N \times 256$	64×256
	SP	–	$M \times 256$	–
	MLP	Linear2	$M \times 512$	256×512
SE	–	Linear3	$M \times 512$	$M \times M$
Decoder	MLP	Linear4	$M \times 64$	512×64
	SUP	–	$N \times 64 \times 1 \times 1$	–
	Conv	Unsample	$N \times 64 \times 7 \times 7$	–
		Conv3	$N \times 64 \times 7 \times 7$	$64 \times 64 \times 3 \times 3$
		Conv4	$N \times D \times 7 \times 7$	$64 \times D \times 1 \times 1$
# Trainable Parameters (w/o SP)				1.234×10^{10}
# Trainable Parameters (w/ SP)				0.287×10^{6}

Note: # Trainable Parameters (w/o SP) denotes the total number of trainable parameters when using a pixel-level DSC on the Salinas dataset, which is computational prohibitive in practice. By using SP, the trainable parameters are reduced by 4 orders of magnitude on the Salinas dataset.

E. Neighborhood Contrastive Regularization

For the clustering task, we expect samples from the same cluster to have high similarity, while samples from different clusters should be as dissimilar as possible. Furthermore, the self-expression coefficient Z_{ij} can be treated as the similarity between samples $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$. Based on this observation, we propose NCR to encourage intraclass similarity and interclass dissimilarity. A schematic representation of the proposed NCR is shown in Fig. 3. Our NCR is motivated by the recent success of contrastive learning to solve similar problems [42]. Nevertheless, unlike the standard contrastive learning approach [43], which sees two augmented views of the same input as positive pairs, we define positive samples using neighborhood relationships. Specifically, given an anchor sample $\boldsymbol{x}_i$, its k nearest neighbors $\boldsymbol{x}_j \in \mathcal{N}_k(\boldsymbol{x}_i)$ is defined as positive samples, where $\mathcal{N}_k$ denotes the set of neighboring samples. We determine $\mathcal{N}_k$ by calculating the Euclidean distances on the initial superpixel-level spectral features. To simplify the training, $\mathcal{N}_k$ is constructed in the preprocessing phase and kept fixed during training. For each anchor, there are k positive pairs and $(N-k-1)$ negative pairs. The goal of NCR is to maximize the similarity between positive pairs. This objective function

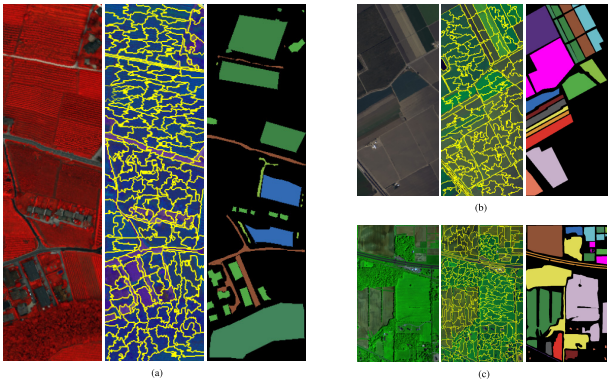


Fig. 4. Visualization of (a) Trento, (b) Salinas, and (c) Indian Pines 2010 datasets. For each dataset, the first, second, and third images denote the false-color image, the homogeneous regions generated by the superpixel segmentation method, and the ground truth map, respectively. The false-color images are composited by the selected bands [40, 20, 10], [25, 19, 8], and [200, 120, 50]. The number of superpixels is set to 227, 171, and 277.

can be defined as the following normalized temperature-scaled cross-entropy loss (NT-Xent) [44]:

$$\mathcal{L}_{\mathrm{NC}} = \frac{1}{M}\sum_{i=1}^{M}\sum_{j\in\mathcal{N}_k(i)} -\log\left(\frac{\exp\left(Z_{ij}/\tau\right)}{\sum_{t=1}^{M}\left[\mathbb{1}_{t\neq i}\exp(Z_{\mathrm{it}}/\tau)\right]}\right) \quad (7)$$

where $\mathbb{1}_{j\neq i} = \{0, 1\}$ is an indicator function and τ denotes a temperature parameter. We set τ to 0.5 for all experiments.

Since there is higher confidence that the neighbors of an anchor have the same semantics and belong to the same subspace, NCR will explicitly attract similar samples and repel dissimilar samples, resulting in a robust subspace representation. Furthermore, NCR acts as a spatial constraint, which can smooth the boundaries of different clusters. There are three significant differences between existing contrastive learning methods and NCR. First, previous methods construct positive and negative pairs based on data augmentation [45], [46], while NCR does not need that. Second, NCR allows each anchor sample to be paired with multiple positive samples. In contrast, most previous methods consider only one positive pair [42]. Third, NCR calculates agreement using self-expression coefficients instead of pairwise similarity. For example, most instance-level contrastive learning methods use cosine similarity.

F. Objective Function and Optimization

Our objective function consists of three losses: pixel-level reconstruction loss $\mathcal{L}_{\mathrm{AE}}$, superpixel-level reconstruction loss $\mathcal{L}_{\mathrm{SE}}$, and nonlocal NCR loss $\mathcal{L}_{\mathrm{NC}}$. Thus, the training procedure entails a typical multitask learning problem. However, tuning the weight of each task by hand is a difficult and expensive process. To avoid this dilemma, we follow the multitask learning principle suggested by Kendall *et al.* [47] and weigh the three losses by estimating the homoscedastic uncertainty of each task. The resulting adaptive loss function is defined as

$$\mathcal{L} = \frac{1}{2\sigma_1^2}\mathcal{L}_{\mathrm{AE}} + \frac{1}{2\sigma_2^2}\mathcal{L}_{\mathrm{SE}} + \frac{1}{2\sigma_3^2}\mathcal{L}_{\mathrm{NC}} + \log\left(\prod_{i=1}^{3}\sigma_i\right) \quad (8)$$

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TABLE III
COMPARISON OF CLUSTERING PERFORMANCE ON THE TRENTO DATASET

Class	k-Means [32]	FCM [33]	LSC [19]	SSSC [18]	ESC [20]	HESSC [17]	S5C [16]	DEC [37]	AE+k-Means	NCSC
Apple trees	0.4343	0.4343	0.5503	0.5223	0.3508	0.6859	0.8042	0.9668	0.9757	0.9378
Buildings	0.6831	0.6231	0.7248	0.0014	0.7344	0.7399	0.5481	0.7282	0.7210	0.1151
Ground	0.0000	0.0000	0.0000	0.0000	0.0209	0.9708	0.0125	0.0000	0.0000	0.0835
Wood	0.4684	0.5203	0.4821	0.9772	0.7365	0.7724	0.4740	0.8410	0.7636	0.9566
Vineyard	0.7133	0.6029	0.6907	0.7011	0.7798	0.4185	0.5743	0.6090	0.5925	0.7115
Rolds	0.7505	0.7766	0.6588	0.7221	0.6434	0.1408	0.6241	0.6430	0.6525	0.7703
ACC	0.5918	0.5661	0.5979	0.6845	0.6787	0.5716	0.5685	0.7322	0.7046	**0.7546**
Kappa	0.4909	0.4677	0.4987	0.5790	0.5720	0.4412	0.4667	0.6564	0.6228	**0.6851**
NMI	0.4701	0.4739	0.4768	0.5027	0.4997	0.4520	0.3974	0.6116	0.5753	**0.7046**
ARI	0.3972	0.3663	0.3858	0.5119	0.4555	0.3028	0.3337	0.5323	0.4697	**0.7061**
Purity	0.6918	0.6884	0.6744	0.7209	0.7063	0.5767	0.6761	0.7399	0.7151	**0.8448**
Running Time	2.03	0.24	3.58	16.97	35.52	644.19	479.55	435.32	400.28	673.30

Note: The best results are shown in bold. The running time (in second) denotes the complete CPU/GPU training time.

TABLE IV
COMPARISON OF CLUSTERING PERFORMANCE ON THE SALINAS DATASET

Class	k-Means [32]	FCM [33]	LSC [19]	SSSC [18]	ESC [20]	HESSC [17]	S5C [16]	DEC [37]	AE+k-Means	NCSC
Brocoli_green_weeds_1	0.0000	0.9278	0.0000	0.0000	0.0000	0.9278	0.0000	0.0000	0.9000	0.0000
Brocoli_green_weeds_2	1.0000	0.1119	0.9949	0.9928	0.9938	0.6887	0.9844	1.0000	0.7933	0.9125
Fallow	0.0000	0.0000	0.5294	0.0486	0.2915	0.0000	0.1204	0.5501	0.0000	0.6538
Fallow_rough_plow	0.0000	0.0000	0.9907	0.7439	0.9964	0.0000	0.9663	0.9340	0.5079	0.0000
Fallow_smooth	0.9907	0.2670	0.8073	0.9093	0.8857	0.9899	0.7976	0.7367	0.7412	0.9985
Stubble	1.0000	0.0000	0.8990	0.9495	0.9194	0.5792	0.8894	0.8406	0.8366	0.9970
Celery	0.6463	0.8055	0.9972	0.5393	0.9486	0.7726	0.8838	0.3328	0.6943	1.0000
Grapes_untrained	0.2914	0.7828	0.2972	0.3352	0.9378	0.5955	0.2700	0.5008	0.8565	0.8568
Soil_vinyard_develop	0.9932	0.9919	0.9763	0.6695	0.9886	0.9321	0.3082	0.9997	0.9994	0.9997
Corn_senesced_green_weeds	0.6367	0.4549	0.3383	0.2477	0.2123	0.1312	0.5018	0.5863	0.3350	0.8005
Lettuce_romaine_4wk	0.0000	0.0000	0.9551	0.6330	0.0000	0.0000	0.3970	0.0000	0.0000	0.0000
Lettuce_romaine_5wk	0.5719	0.5864	0.9995	0.6440	0.8895	0.0000	0.3752	0.9460	0.8293	0.3456
Lettuce_romaine_6wk	0.9967	0.0000	0.0000	0.0000	0.0011	0.9214	0.0000	0.0000	0.0000	0.0000
Lettuce_romaine_7wk	0.9794	0.0009	0.0000	0.2748	0.9607	0.8131	0.2794	0.9645	0.0000	0.0000
Vinyard_untrained	0.4199	0.2369	0.8013	0.4330	0.0003	0.1560	0.2720	0.6237	0.0579	0.9996
Vinyard_vertical_trellis	0.9397	0.7227	0.9286	0.9065	0.8866	0.4079	0.8379	0.9917	0.9092	1.0000
ACC	0.5911	0.4897	0.6722	0.5301	0.6802	0.5293	0.4731	0.6568	0.6257	**0.7965**
Kappa	0.5541	0.4206	0.6398	0.4868	0.6385	0.4785	0.4318	0.6206	0.5796	**0.7729**
NMI	0.7662	0.6085	0.7902	0.6348	0.7938	0.6238	0.6186	0.7720	0.7705	**0.8512**
ARI	0.5167	0.4006	0.5896	0.4111	0.6185	0.4150	0.3738	0.5548	0.5824	**0.7832**
Purity	0.7259	0.4957	0.7244	0.6362	0.6981	0.5881	0.6324	0.7272	0.6917	**0.7965**
Running Time	3.16	0.68	16.61	113.47	245.08	667.59	868.56	817.73	750.06	936.78

where $\{\sigma_i\}_{i=1,2,3}$ indicates the trainable variance of each task. When σ_i becomes larger, which means that the uncertainty of the task is increased, the corresponding task weights decrease proportionally. We jointly optimize the model parameters and these variances using an adaptive moment estimation (Adam) optimizer in an end-to-end fashion.

IV. RESULTS

A. Experiment Setup

1) Datasets: We conduct experiments on three complete large HSI datasets: Trento, Salinas, and Indian Pines 2010. The Trento dataset was captured by the AISA Eagle sensor over a rural area south of the city of Trento, Italy. The dataset is composed of 600 $\times$ 166 pixels with a spatial sampling of 1 m and contains 63 spectral bands ranging between 0.40 and 0.98 μ m. There are six ground truth classes in the dataset. The Salinas dataset was collected by the 224-band AVIRIS sensor over the Salinas Valley, California. After removing water absorption and noise from the bands covering the region, the number of spectral bands used in experiment is 204. There are 16 land cover types and 512 $\times$ 217 pixels in the dataset. The Indian Pines 2010 image was captured from the ProSpecTIR system in the neighborhood of Purdue University, Indiana, in May 2010. The dataset consists of 360 bands, sprawled over spatial dimensions of 445 $\times$ 750, and 16 different classes of land covers. In Fig. 4, we provide the visualization with respect to the false-color image, homogeneous regions, and ground truth of these datasets. The detailed land cover types of these datasets are described in Table I.

2) Implementation Details: We implement our method according to the parameters provided in Table II. In the encoder, we use two convolutional layers with kernels of 1 $\times$ 1 $\times$ 64 and 3 $\times$ 3 $\times$ 64, and a linear layer with 256 neurons before the SP layer and 512 neurons after the SP layer, each followed by a ReLU and batch normalization layer. Our model is trained on a full batch HSI patches, each with the patch size of 7 $\times$ 7. The initial learning rate is set to 0.025 for the SE layer and 0.0005 for other layers. During the training phase, the learning rates are divided by ten after every 50 epochs. The weight decay parameter is set to 5×10^{-4} except for the SE layer. We train the model with 200 epochs in total for

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TABLE V
COMPARISON OF CLUSTERING PERFORMANCE ON THE INDIAN PINES 2010 DATASET

Class	k-Means [32]	FCM [33]	LSC [19]	SSSC [18]	ESC [20]	HESSC [17]	S5C [16]	DEC [37]	AE+k-Means	NCSC
Corn-high	0.9652	0.0000	0.8981	0.7668	0.8837	0.8772	0.7555	0.7629	0.8955	0.0000
Corn-mid	0.1862	0.0000	0.0000	0.1368	0.0966	0.0954	0.2322	0.1011	0.0213	0.0000
Corn-low	0.0000	0.0000	1.0000	0.9972	0.9775	0.9775	0.9129	0.9831	0.0000	0.0000
Soy-bean-high	0.0000	0.0000	0.0000	0.0073	0.0000	0.0000	0.0037	0.0000	0.0000	0.0000
Soy-bean-mid	0.3256	0.1812	0.3691	0.3980	0.3927	0.3871	0.2720	0.2494	0.5324	0.8563
Soy-bean-low	0.3998	0.8931	0.4308	0.3672	0.4257	0.5818	0.2218	0.3562	0.4062	0.9008
Residues	0.5408	0.0000	0.4469	0.3349	0.3993	0.3705	0.2280	0.4443	0.1287	1.0000
Wheat	0.7080	0.1721	0.7189	0.4317	0.6888	0.1878	0.3094	0.2755	0.2554	0.6888
Hay	0.5155	0.8172	0.4740	0.5191	0.7949	0.8691	0.4673	0.8440	0.4804	0.8730
Grass/Pasture	0.2796	0.3954	0.9830	0.4540	0.0795	0.3773	0.6347	0.5292	0.6605	0.9969
Cover crop 1	0.9155	0.2072	0.9789	0.7054	0.5299	0.0000	0.9264	0.9873	0.9614	1.0000
Cover crop 2	1.0000	0.0000	0.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.0000
Woodlands	0.3291	0.3948	0.7168	0.7342	0.7051	0.5652	0.9093	0.4831	0.6261	0.2393
Highway	1.0000	0.6105	1.0000	0.8643	0.8709	0.9371	1.0000	0.6058	0.5219	0.8974
Local road	0.0000	0.0000	0.0000	0.0000	0.9861	0.5418	0.0000	0.0000	0.6036	0.2271
Buildings	0.0147	0.1832	0.0000	0.0000	0.0330	0.0458	0.0000	0.4945	0.0989	0.3278
ACC	0.4342	0.5020	0.5377	0.5294	0.5972	0.5956	0.5204	0.5424	0.5176	**0.6821**
Kappa	0.3799	0.4152	0.4808	0.4670	0.5329	0.5158	0.4614	0.4777	0.4560	**0.6349**
NMI	0.6267	0.4716	0.6852	0.6361	0.6638	0.5596	0.6097	0.6334	0.6827	**0.7305**
ARI	0.3782	0.3434	0.4962	0.4933	0.5596	0.4325	0.5394	0.5134	0.4926	**0.6588**
Purity	0.8119	0.6283	0.8365	0.8299	0.8328	0.6918	0.7886	0.7800	0.8388	**0.8683**
Running Time	22.76	2.75	147.07	632.69	967.26	1566.73	1601.78	2751.61	2557.47	2915.13

TABLE VI
EVALUATION OF THE EFFECTIVENESS OF THE PROPOSED NCR

Regularizer	Trento					Salinas					Indian Pines 2010				
	ACC	Kappa	NMI	ARI	Purity	ACC	Kappa	NMI	ARI	Purity	ACC	Kappa	NMI	ARI	Purity
Baseline	0.4576	0.3128	0.2905	0.2533	0.6051	0.5408	0.4977	0.6274	0.3847	0.5924	0.3412	0.2842	0.3892	0.1886	0.6066
w/ ℓ_1	0.6017	0.4921	0.4870	0.4634	0.7369	0.5392	0.4960	0.6316	0.4225	0.5956	0.4011	0.3327	0.4161	0.2700	0.6373
w/ ℓ_2	0.5620	0.4563	0.3896	0.3760	0.6455	0.6041	0.5590	0.7020	0.5593	0.6271	0.3561	0.2995	0.3804	0.1789	0.6082
w/ GR	0.6418	0.5357	0.4770	0.4396	0.6695	0.6706	0.6301	0.7975	0.6494	0.6885	0.5203	0.4578	0.5545	0.4175	0.7479
w/ NCR	**0.7546**	**0.6851**	**0.7046**	**0.7061**	**0.8448**	**0.7965**	**0.7729**	**0.8512**	**0.7832**	**0.7965**	**0.6821**	**0.6349**	**0.7305**	**0.6588**	**0.8683**

Note: Baseline: no regularization; ℓ_1: ℓ_1-norm; ℓ_2: ℓ_2-norm; GR: graph regularization; NCR: neighborhood contrastive regularization.

TABLE VII
COMPARISON OF CLUSTERING PERFORMANCE WITH SUPERPIXEL-LEVEL FEATURE REPRESENTATION

Dataset	Metric	k-Means [32]	FCM [33]	LSC [19]	SSSC [18]	ESC [20]	HESSC [17]	S5C [16]	DEC [37]	AE+k-Means	EKGCSC [11]	NCSC
Trento	ACC	0.6560	0.5263	0.7388	0.6679	0.5906	0.6084	0.6618	0.7365	0.6704	0.6629	**0.7546**
	Kappa	0.5216	0.3637	0.6589	0.5742	0.4757	0.4285	0.5661	0.6434	0.5514	0.5355	**0.6851**
	NMI	0.5389	0.3189	0.6142	0.5908	0.4447	0.4141	0.5306	0.6049	0.6091	0.5437	**0.7046**
	ARI	0.5100	0.2994	0.6090	0.6223	0.3446	0.3066	0.5325	0.6214	0.5581	0.4990	**0.7061**
	Purity	0.6919	0.6010	0.7588	0.7594	0.6659	0.6130	0.7789	0.7409	0.7308	0.7136	**0.8448**
Salinas	ACC	0.6475	0.6404	0.5697	0.6809	0.4392	0.4335	0.7418	0.6875	0.6751	0.7238	**0.7965**
	Kappa	0.5994	0.6048	0.5223	0.6487	0.3960	0.3102	0.7085	0.6458	0.6314	0.6872	**0.7729**
	NMI	0.7808	0.7506	0.7223	0.8239	0.5333	0.4739	**0.8637**	0.8187	0.7922	0.8486	0.8512
	ARI	0.5900	0.5256	0.4275	0.6200	0.2550	0.2935	0.6857	0.6256	0.6095	0.6522	**0.7832**
	Purity	0.6572	0.7220	0.6384	0.7036	0.5342	0.5573	0.7520	0.6995	0.6751	0.7332	**0.7965**
Ind. Pines	ACC	0.5962	0.4558	0.5086	0.5040	0.4107	0.4402	0.5261	0.6243	0.6138	0.6470	**0.6821**
	Kappa	0.5246	0.3931	0.4445	0.4508	0.3590	0.3383	0.4718	0.5664	0.5377	0.5912	**0.6349**
	NMI	0.5996	0.5737	0.5951	0.6304	0.5558	0.5633	0.6422	0.6899	0.6015	0.6961	**0.7305**
	ARI	0.4656	0.3567	0.4133	0.4064	0.3023	0.2202	0.4199	0.5763	0.4944	0.5931	**0.6588**
	Purity	0.7421	0.7521	0.7638	0.7997	0.7575	0.4708	0.7934	0.8207	0.7499	0.8409	**0.8683**

all datasets. For computational efficiency, we reduce spectral bands into eight using the principal component analysis. The training is conducted on an NVIDIA TITAN XP GPU with 11 GB of graphic memory. More implementation details can be found in our released source code.[2]

3) Baseline Methods: We compare our method with nine representative scalable clustering approaches: k-Means [32], FCM [33], landmark-based spectral clustering (LSC) [19], scalable SSC (SSSC) [18], exemplar-based subspace clustering (ESC) [20], hierarchical sparse subspace clustering (HESSC) [17], selective sampling-based scalable sparse subspace clustering (S5C) [16], DEC [37], and AE + k-Means. We follow the settings suggested in the corresponding articles. The number of landmarks (or exemplars/subsamples) is set to $(200 \times C)$, $(200 \times C)$, 100, and $(200 \times C)$ for LSC, SSSC, ESC, and S5C, respectively. For DEC, we first pretrain

[2]https://github.com/AngryCai/NCSC

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a denoising AE with a learning rate of 0.001 for 100 epochs and then fine-tune the encoder via a clustering loss by setting the learning rate and epoch to 100 and 0.00002, respectively. In AE + k-Means, we use k-Means to generate clustering results on the features extracted by a trained AE, which removes the SP, SE, and SUP layers of SPAE.

4) Evaluation Metrics: Five widely used and relevant metrics are used to quantitatively evaluate the clustering performance: overall accuracy (ACC), Kappa coefficient (Kappa), normalized mutual information (NMI), adjusted Rand index (ARI), and Purity [7], [16], [27]. These metrics range in [0, 1] (ACC, NMI, and Purity) or $[-1, 1]$ (Kappa and ARI), and higher scores signify that more accurate clustering results have been achieved. Following the common process of clustering evaluation, we adopt the Hungarian algorithm [48] to match the predicted labels to the ground truth before calculating metrics.

B. Comparison With State-of-the-Art Methods

In Tables III–V, we report the numerical clustering results on the Trento, Salinas, and Indian Pines 2010 datasets, respectively. For fair comparisons among different methods, we reproduce baselines under the same data preprocessing as NCSC. We reach the following two conclusions. First, NCSC outperforms all other baseline methods by a significant margin on all datasets. For example, NCSC achieves at least 2.24, 11.63, and 8.49 percent points of improvement over baselines on the Trento (ACC = 75.46%), Salinas (ACC = 79.65%), and Indian Pines 2010 (ACC = 68.21%) datasets, respectively. This attests for the effectiveness of the proposed method. Second, NCSC and other deep clustering models tend to spend more running time because of many trainable parameters. The time complexity of training NCSC one epoch can be roughly denoted as $O(\mathcal{E}) + O(M^2) + O(\mathcal{D})$. The process requires 314.82×10^9 floating-point operations (FLOPs) on the Trento dataset, which is higher than DEC (178.433×10^9 FLOPs). In contrast, k-Means and FCM need linear time complexity of $O(N)$, while the spectral clustering-based methods (e.g., LSC, SSCC, and ESC) need at least cubic complexity of $O(p^3)$, where p is the number of anchor points. Despite higher time complexity, deep clustering models are significantly better than shallow models, which mainly benefit from their high-level feature extraction mechanisms. We also find that DEC is more competitive than AE + k-Means since DEC can learn a cluster-friendly feature representation. Third, scalable spectral/subspace clustering approaches (i.e., LSC, SSSC, ESC, HESSC, and S5C) are helpful for scaling up to large HSI datasets, but they consistently ignore contextual information in HSI. Our NCSC scales up subspace clustering to large images through SP, which results in more efficient use of context, thus yielding a more competitive clustering result. It should be noted that classic spectral/subspace clustering models and DSC failed to handle the three datasets because of memory saturation. To summarize, the proposed NCSC approach not only enables DSC to deal with large HSI images but also achieves performance improvement over existing clustering methods.

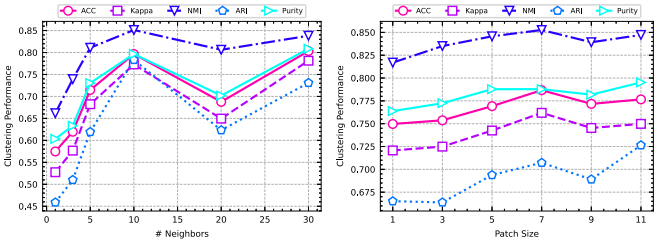


Fig. 5. (a) Impact of the number of neighbors of NCR on the Salinas dataset (accuracy versus neighbor size). (b) Influence of patch size on the Salinas dataset (accuracy versus patch size).

C. Ablation Study

1) Importance of NCR: We compare different regularizations in Table VI. For a fair comparison, we train the same SPAE architecture while varying the regularization used in the SE layer. Specifically, we assess the superiority of ℓ_1-norm, ℓ_2-norm, GR, and NCR. We treat the model without regularization as the baseline. Following our previous work [27], we construct a kNN graph for GR by setting k to the same as for NCR. When no regularization is used, the clustering performance of the baseline is dramatically decreased because of the trivial solution. This confirms, as expected, that regularization is crucial to DSC architecture. Our proposed NCR outperforms the baseline by the largest margin (ACC +29.70%, +25.57%, and +34.09% on the Trento, Salinas, and Indian Pines 2010, respectively) compared with the other three regularizations, demonstrating the effectiveness of the proposed NCR. Furthermore, both NCR and GR achieve significantly better clustering performance than the ℓ_1-norm and ℓ_2-norm. A conceivable reason is that context is a more robust inductive bias than sparse prior for HSI. Although GR shows promising results, NCR yields up to 11.28, 16.18, and 12.59 percentage points in improvement over GR, demonstrating the superiority of NCR.

2) Comparison With Superpixel-Level Results: Here, we explore the impact of superpixel segmentation on the clustering performance. To this end, we compare the baselines under the superpixel-level feature representation transformed by (2). This setting also makes it possible to apply those methods that are memory prohibitive and computationally expensive. Thus, we compare our approach to a state-of-the-art model, the efficient kernel graph convolutional subspace clustering (EKGCSC) [11]. We report the comparative results in Table VII. It can be seen that the proposed NCSC achieves the best clustering accuracy. Compared to EKGCSC which incorporates graph convolution and a kernel trick, NCSC delivers noticeable accuracy improvements. Specifically, the margins of ACC between NCSC and EKGCSC are 9.17, 7.27, and 3.51 percentage points on the Trento, Salinas, and Indian Pines 2010 datasets, respectively. In addition, some of the compared baselines (e.g., k-Means, S5C, and DEC) tend to be significantly improved by the superpixel-level representation compared with the pixel-level results. This indicates that the use of the intrinsic homogeneity of HSI not only reduces computational complexity but also has the potential to improve the clustering performance. However, the

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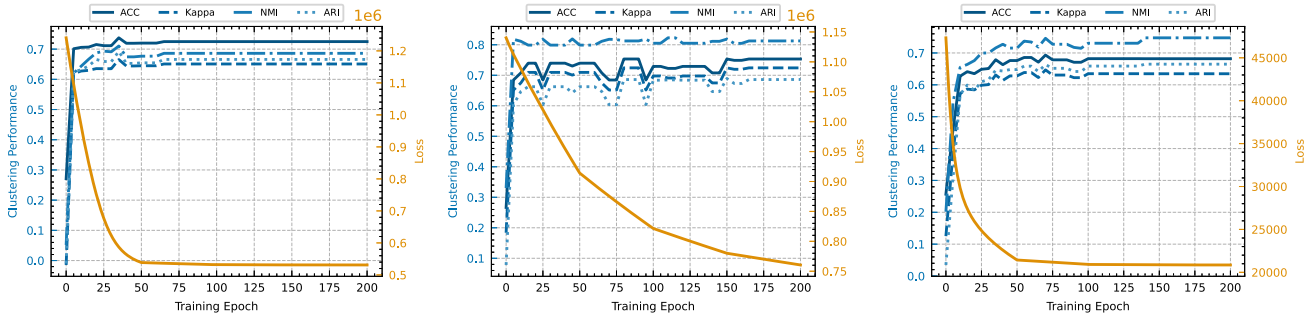


Fig. 6. Clustering performance of NCSC with increasing epochs on (a) Trento, (b) Salinas, and (c) Indian Pines 2010 datasets, where the x-axis denotes the training epoch, and the left and right y-axis denote the clustering metrics and the corresponding loss values, respectively.

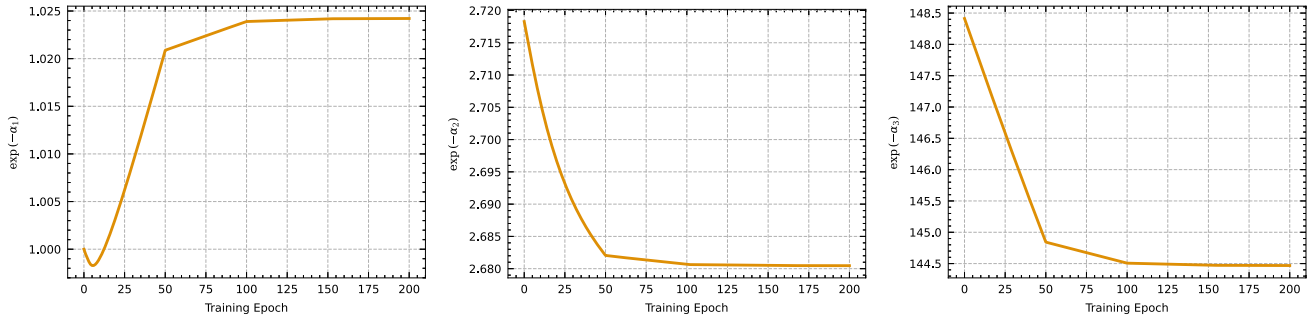


Fig. 7. Evolution of loss weights over training epochs on the Indian Pines 2010 dataset, where the loss weight is indicated as $\exp(-\alpha_i)$ with $\alpha_i = \log(\sigma_i^2)$. (a) Weight of $\mathcal{L}_{\mathrm{AE}}$. (b) Weight of $\mathcal{L}_{\mathrm{SE}}$. (c) Weight of $\mathcal{L}_{\mathrm{NC}}$.

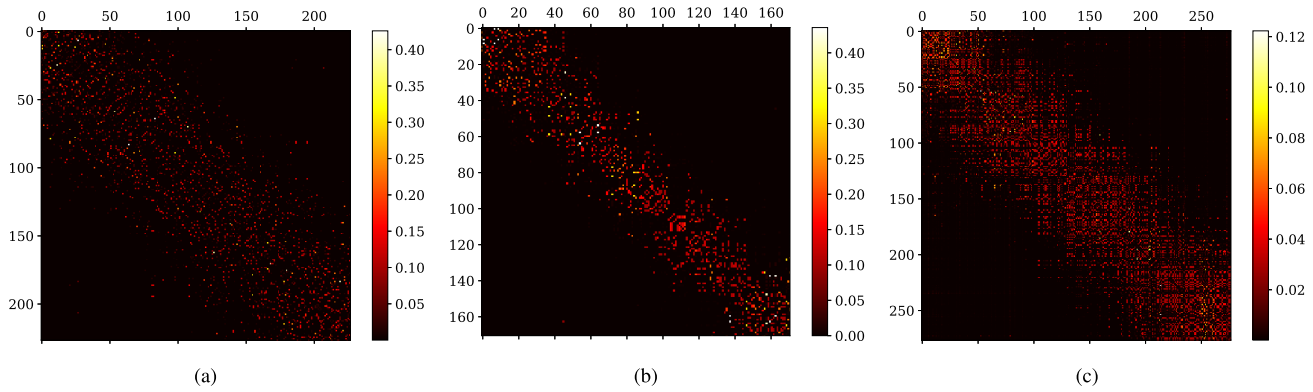
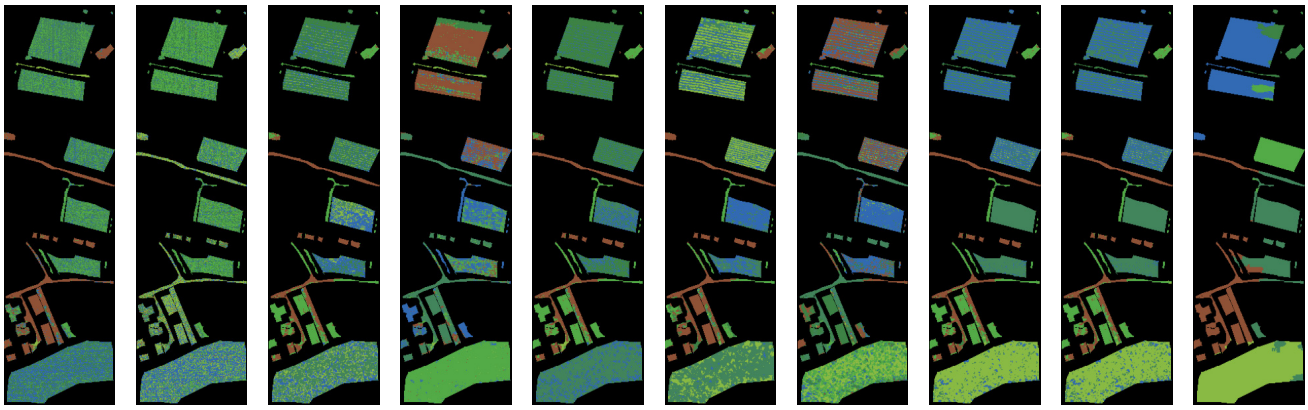


Fig. 8. Visualization of the self-expression coefficients obtained by NCSC. (a) Trento. (b) Salinas. (c) Indian Pines 2010.

superpixel-level representation may also decrease clustering performance because of the following twofold reason. First, the clustering results of different clustering methods are highly dependent on the accuracy of superpixel segmentation and the number of superpixels. Second, these baselines except for EGCSC fail to explore the neighborhood information, leading to poorer spatial consistency.

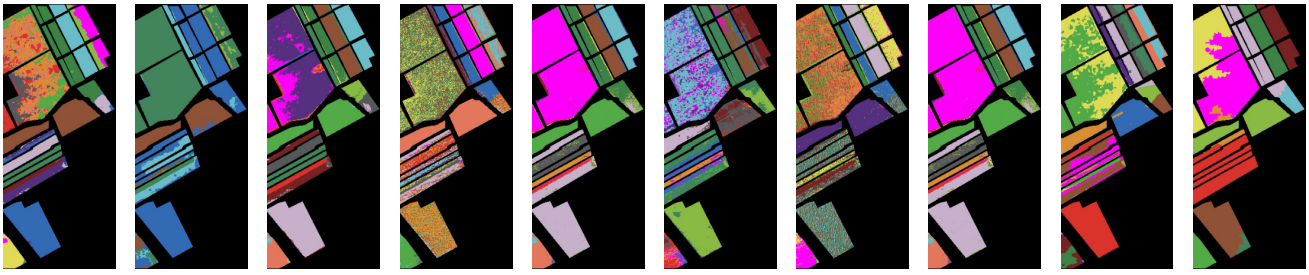
3) Impact of Neighborhood: In Fig. 5(a), we compare the clustering performance of NCSC under a varying number of neighbors in NCR. We vary the parameter k in $\{1, 3, 5, 10, 20, 30\}$ and calculate the clustering metrics on the Salinas dataset. It can be observed that NCSC tends to reach better clustering performance with an increasing number of neighbors. For example, the ACC of NCSC is improved from 57.47% to 80.33% (+22.86 percentage points) when k is increased from 1 to 30. This is due to the fact that the neighborhoods of a sample usually belong to the same subspace and thus benefit the robustness of self-expression. We note that the definition of NCR [see (7)] encourages neighborhoods to be brought close to each other, while other samples are pushed away. This makes the learned subspace representation robust to noise samples. In general, false neighborhoods are allowed by NCR because their contribution during the reconstruction will be limited by true neighborhoods.

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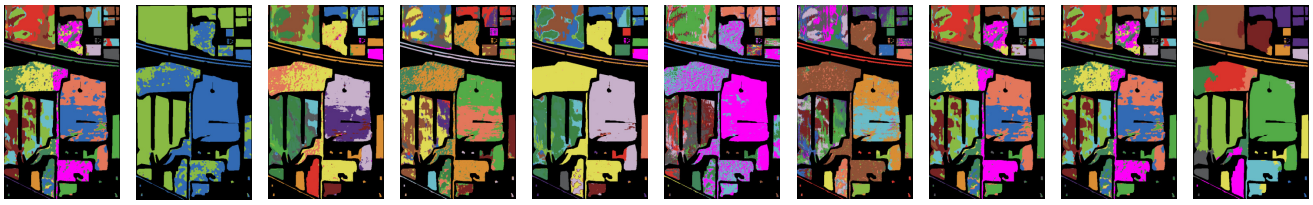
(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 9. Clustering maps for the Trento dataset obtained by different clustering approaches. (a) k-Means. (b) FCM. (c) LSC. (d) SSSC. (e) ESC. (f) HESSC. (g) S5C. (h) DEC. (i) AE + k-Means. (j) NCSC.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 10. Clustering maps for the Salinas dataset obtained by different clustering approaches. (a) k-Means. (b) FCM. (c) LSC. (d) SSSC. (e) ESC. (f) HESSC. (g) S5C. (h) DEC. (i) AE + k-Means. (j) NCSC.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 11. Clustering maps for the Indian Pines 2010 dataset obtained by different clustering approaches. (a) k-Means. (b) FCM. (c) LSC. (d) SSSC. (e) ESC. (f) HESSC. (g) S5C. (h) DEC. (i) AE + k-Means. (j) NCSC.

However, when k becomes large enough [e.g., $k > 10$ in Fig. 5(a)], the probability of including false neighborhoods will increase greatly, leading to an unstable model. This means that there will be an optimal setting for parameter k on different datasets. In our experiment, we set the value of k to 10 empirically.

4) Influence of Patch Size: Fig. 5(b) shows the effect of the patch sizes of inputs on clustering performance. We increase the patch size from 1 to 11 with an interval of 2. We can observe that a larger patch size improves the clustering performance of NCSC. This tendency is consistent with the characteristic of HSI. Generally, an HSI patch with proper size [e.g., 7×7 in Fig. 5(b)] can greatly enhance the discriminative property of the central pixel, thus benefiting the exploration of the subspace. However, it is not recommended to use a too large patch size because it will introduce an increased risk of mixing classes, as well as a large computational burden.

5) Convergence Analysis: To analyze the convergence of the NCSC model, we visualize clustering metrics against loss values of 200 training epochs in Fig. 6(a)–(c). We use a matrix of ones to initialize the SE layer. As a result, the initial clustering accuracy of NCSC is extremely poor. After several training epochs, we can observe that clustering metrics are significantly increased and then become stable, whereas the loss values tend to be convergent. This demonstrates that NCSC can gradually improve the quality of the subspace representation by jointly training. In practice, we note that an appropriate initialization of the SE layer (e.g., using an adjacent matrix of a graph [14]) would further accelerate model training.

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In Fig. 7(a)–(c), we show the evolution of the proposed adaptive loss weights on the Indian Pines 2010 dataset. In order to avoid numerical instability during training, we predict the log variance, $\alpha_i = \log(\sigma_i^2)$, instead of σ_i. Thus, we derive the loss weight, $\exp(-\alpha_i)$. The initial value of α_i is set to $\alpha_1 = 0$, $\alpha_1 = -1$, and $\alpha_1 = -5$. This leads to a larger NCR weight, and thus, the model will pay more attention to optimizing the subspace representation coefficients. We find that these loss weights are dynamically updated with the training epoch. At the conclusion of training, the three tasks have uncertainty estimates of 0.9764, 0.3731, and 0.0069, which results in effective weighting between the tasks of $1.02 : 2.68 : 144.47$. We find a similar tendency on the other two datasets. This demonstrates that the SE layer, especially the proposed NCR, plays a more important role during optimization.

D. Visualization

1) Visualization of Self-Expression Coefficients: In Fig. 8, we visualize the obtained self-expression coefficient matrix of NCSC. For better visualization, we highlight the nonzero elements of $\mathbf{W}$ by sharpening with the third power of $\mathbf{W}$, and each row of the matrix is normalized. We can observe that the obtained coefficient matrices showcase good sparsity and structural properties. In particular, the nonzero values are explicitly restricted within the neighboring regions by NCR. It should be noted that the initial coefficient matrix is dense and meaningless (i.e., a matrix of ones). This signifies that NCSC is capable of capturing inherent subspace relationships among samples.

2) Visualization of Clustering Maps: Figs. 9–11 show the qualitative comparison of clustering maps predicted by different clustering approaches on the Trento, Salinas, and Indian Pines 2010 datasets, respectively. Due to the absence of context information, baseline methods (e.g., k-Means, FCM, and LSC) yield poorer clustering maps with obvious salt-and-pepper noises. This issue can be alleviated by using the smoothing technique at a postprocessing stage or considering spatial prior during modeling. In contrast, our NCSC can generate cleaner clustering maps with desirable homogeneity on all datasets, especially in terms of texture and edge details. This is because our method is dually constrained by local homogeneous regions and nonlocal neighborhood relationships. Furthermore, high-level and nonlinear features make it easier for NCSC to discover the underlying subspace, thereby achieving a more accurate clustering map.

V. Conclusion

In this article, we devote attention to scaling up the DSC model to large HSI. Specifically, we propose an SPAE to efficiently learn contracted subspace representations at the superpixel level, which significantly relaxes both the time and space complexity of self-expression. In addition, we introduce an NCR, which explicitly attracts nonlocal neighborhood and repels dissimilar samples in the subspace, thus effectively improving the robustness of subspace representations. We adopt an adaptive multitasks learning scheme, which weights different losses by predicting the homoscedastic uncertainty of each task, to jointly train the resulting model. Experiments on three hyperspectral datasets show the remarkable superiority of our method over state-of-the-art clustering methods.

Our successful attempt shows that the use of local spatial homogeneity of HSI not only overcomes the self-expression bottleneck in DSC but also improves accuracy and efficiency. Furthermore, we also observe that NCR will not only significantly improve model robustness but also can be used as a general regularization method for existing subspace models. We note that the superpixel association and neighborhood are precomputed and kept fixed during training. Thus, future work may include incorporating an end-to-end superpixel segmentation network (e.g., FCN [41]) and exploring the dynamic neighborhood in latent space using graph convolutional networks [49].

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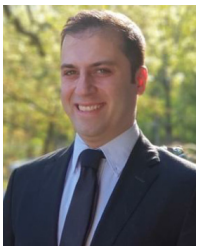
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